

3. Phase 3: Unsupervised Phenotype Stratification (N = 699)

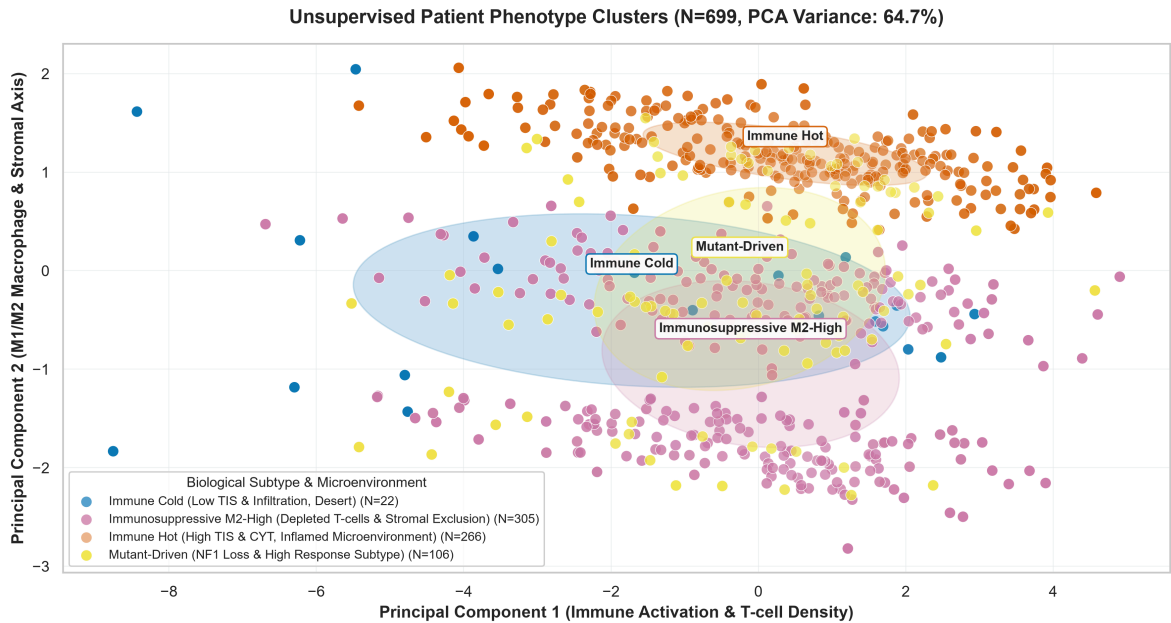
 **Analytical Methodology & Rationale**

- What is being done:** Applying Gaussian Mixture Models (GMM, $K = 4$) with full covariance matrices to feature matrices, calculating soft posterior probabilities, and generating 2D Principal Component projections.
- Why we are doing it:** Unsupervised clustering discovers natural biological patient subgroups without outcome bias. Soft probabilistic GMM clustering accommodates non-spherical correlated feature distributions and quantifies patient membership uncertainty.
- What question it answers:** What distinct patient clusters emerge from multi-dimensional biological profiling, and how are patients soft-partitioned across biological phenotypes?

Unsupervised Phenotype Cluster Summary

Cluster ID	Biological Phenotype Subtype	Patient Count (N)	Cohort Share	Response Rate
Cluster 0	Immune Cold (Low TIS & Infiltration, Desert)	22	3.1%	66.7%
Cluster 1	Immunosuppressive M2-High (Depleted T-cells & Stromal Exclusion)	305	43.6%	37.1%
Cluster 2	Immune Hot (High TIS & CYT, Inflamed Microenvironment)	266	38.1%	37.7%
Cluster 3	Mutant-Driven (NF1 Loss & High Response Subtype)	106	15.2%	60.7%

Unsupervised Phenotype Cluster Projection (2D PCA)



 **Figure Interpretation: 2D Principal Component Cluster Projection**

- What this plot shows:** 2D Principal Component Projection of $N = 699$ patients colour-coded by their multi-modal GMM phenotype cluster ($K = 4$). Shaded confidence ellipses mark cluster boundaries.
- Axis 1 (Horizontal):** Principal Component 1 captures immune activation and lymphocytic T-cell density (separating Inflamed Hot vs Desert Cold tumours).
- Axis 2 (Vertical):** Principal Component 2 captures macrophage polarisation (M1/M2 ratio) and stromal CAF exclusion.
- Clinical Value:** Discovers discrete patient subgroups with distinct treatment response profiles without relying on biased outcome labels.

Unsupervised Phenotype Manifold (t-SNE Projection)

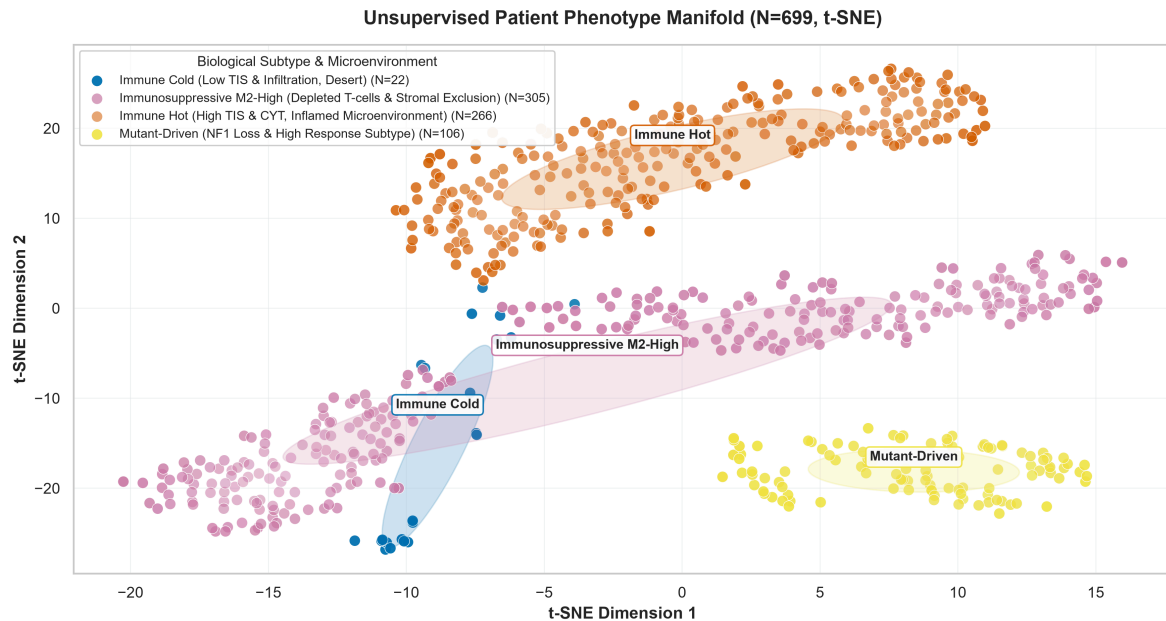


Figure Interpretation: Non-Linear t-SNE Cluster Manifold

- **What this plot shows:** 2D t-SNE non-linear manifold projection of the 9-feature patient space ($N = 699$), colour-coded by the GMM cluster labels assigned in full 9-dimensional feature space.
- **Non-Linear Topology:** t-SNE (perplexity=50) preserves local patient neighbourhood structure and non-linear biomarker interactions across the 9 multi-modal clustering features (TIS , CYT , CD8 T-cells, M1/M2 Macrophages, CAFs, BRAF / NRAS / NF1 mutations). Natural within-cluster scatter reflects genuine continuous variation within each immune phenotype.

Key Takeaways & Student Summary

- **Distinct Patient Groups:** GMM soft clustering partitioned $N = 699$ patients (full cohort) into four biological subgroups; within the ICI-treated sub-cohort ($N = 326$), response rates range from **37.1% to 66.7%**.
- **Highest Response Group:** The **Immune Cold (Low TIS & Infiltration, Desert)** subgroup achieves the highest response rate (66.7%), benefiting from favorable immune activation and high driver mutation burden.
- **Treatment-Resistant Subgroup:** The **Immunosuppressive M2-High (Depleted T-cells & Stromal Exclusion)** subgroup exhibits the lowest response rate (37.1%), highlighting the need for targeted combination therapies beyond single-agent PD-1 blockade.

Student-Friendly Phase 3 Summary

Phase 3 performed unsupervised multi-dimensional GMM soft clustering to discover natural biological patient subgroups without relying on outcome labels:

1. **Four Distinct Phenotypes:** Gaussian Mixture Models ($K = 4$) partitioned patients into *Immune Hot (High TIS & CYT, Inflamed Microenvironment)*, *Immune Cold (Low TIS & Infiltration, Desert)*, *Immunosuppressive M2-High (Depleted T-cells & Stromal Exclusion)*, and *Mutant-Driven (NF1 Loss & High Response Subtype)* phenotypes across 9 biomarker axes.
2. **Soft Probabilistic Assignments:** Full covariance matrices (Σ_k) accommodate non-spherical feature correlation and calculate continuous posterior membership probabilities $\bar{P}_i$.
3. **Dimensionality Projections:** 2D PCA and non-linear t-SNE projections confirm clear spatial separation, with PC1 capturing T-cell inflammation and PC2 capturing myeloid/stromal exclusion.
4. **Clinical Takeaway:** Identifying a patient's biological phenotype and probability profile provides the foundation for targeted routing rather than applying a single uniform treatment protocol.

Phase 3 Script Execution & Software Module Architecture

- **Primary Pipeline Execution Scripts:**

- `03_cluster_patients.py`: Executes Gaussian Mixture Model (GMM) soft clustering ($K = 4$, full covariance) across the 9 multi-modal feature space, computes posterior probabilities, exports `gmm_posterior_probabilities.csv` and `patient_clusters.csv`, serialises the fitted model (`gmm_model.pkl`), and generates PCA/t-SNE 2D projections (`pca_clusters.png`, `tsne_clusters.png`).
- `08_compare_clustering_algorithms.py`: Benchmarks alternative clustering algorithms (K-Means, HAC, GMM, Spectral Clustering, DBSCAN, Consensus Clustering) across Silhouette, Calinski-Harabasz, Davies-Bouldin, ARI, and response rate spread.
- `09_run_consensus_clustering.py`: Executes 1,000-bootstrap Consensus Clustering ensemble across patients (80%) and features (80%) for $K \in [2, 8]$, generating Consensus CDF curves (`consensus_cdf_curves.png`), Delta Area elbow plots ($\Delta(4) = 0.1499$, `consensus_delta_area.png`), and co-association heatmaps (`consensus_heatmap_k4.png`).

- **Core Supporting Python Modules:**

- `clustering.py`: Implements feature scaling, GMM soft clustering (`run_gmm`), and 2D cluster projection plotting (`plot_2d_cluster_projection`) for PCA and t-SNE manifolds.
- `phenotyping.py`: Implements cluster profiling (`profile_clusters`) and phenotype label mapping (`assign_phenotype_labels`).
- `q5_constants.py`: Central source of truth defining the 9 multi-modal clustering features (`CLUSTERING_FEATURES`) and biological phenotype mappings (`PHENOTYPE_LABEL_MAP`).

- **Shared Cross-Question & Pipeline Modules:**

- `run_q5_pipeline.py`: Master pipeline orchestrator executing `03_cluster_patients.py` as Step 3.
- `generate_q5_report.py`: Reads clustering metrics and updates phase markdown reports.